

Computer Science & IT

Discrete Mathematics

Combinatorics

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic

Formation of recurrence relation



Topic

Solution of recurrence relation using substitution method



Topics to be Covered



Topic

Solution of recurrence relation using method of characteristic roots





Topic : Solution of recurrence relation

- 1) Substitution method ✓
- ✧ 2) Method of characteristic roots
- 3) Method of undetermined coefficients
- 4) Using generating functions



Topic : Method of characteristic roots

Consider the following linear recurrence relation.

$$\lambda_0 a_n + \lambda_1 a_{n-1} + \lambda_2 a_{n-2} + \dots + \lambda_k a_{n-k} = f(n) \quad \text{--- eqn ①}$$

{ Where, $\lambda_0, \lambda_1, \lambda_2, \dots, \lambda_k$ }
are constants

Put $n-k = n$, i.e. $n = n+k$

∴ Equation becomes

$$\lambda_0 a_{n+k} + \lambda_1 a_{n+k-1} + \lambda_2 a_{n+k-2} + \dots + \lambda_k a_{n+k-k} = F(n)$$

if we put $n = n+k$ in f ,
then we get
a new function
of n let's call it

--- eqn ②

Consider the following Shift operator

$$a_{n+i} = E^i a_n$$

i.e. $a_n = a_n \left\{ a_{n+0} = \cancel{E^0} \cdot a_n^1 \right\}$

$$a_{n+1} = E^1 a_n$$

$$a_{n+2} = E^2 a_n$$

$$a_{n+3} = E^3 a_n$$

$$\vdots$$
$$a_{n+k} = E^k a_n$$

$\Rightarrow$ let $a_n = t^n$
 $\therefore a_{n+k} = t^{n+k} = t^n \cdot t^k$
 $= a_n \cdot t^k$
 $= t^k \cdot a_n$

Our eqⁿ ② is

$$\lambda_0 Q_{n+k} + \lambda_1 Q_{n+k-1} + \lambda_2 Q_{n+k-2} + \dots + \lambda_k Q_n = F(n) \text{ --- eqⁿ ②}$$

Put the values from shift operator in eqⁿ ②

$$\lambda_0 E^k Q_n + \lambda_1 E^{k-1} Q_n + \lambda_2 E^{k-2} Q_n + \dots + \lambda_k Q_n = F(n)$$

$$Q_n [\lambda_0 E^k + \lambda_1 E^{k-1} + \lambda_2 E^{k-2} + \dots + \lambda_k] = F(n)$$

it is a polynomial of degree = k
lets call it $\phi(E)$

$$\therefore Q_n \cdot \phi(E) = F(n) \text{ --- eqⁿ ③}$$

$$\rightarrow \lambda_0 a_n + \lambda_1 a_{n-1} + \lambda_2 a_{n-2} + \dots + \lambda_k a_{n-k} = f(n)$$

$$\text{Put } n-k=n, \text{ i.e. } n=n+k$$

if $f(n)=0$, then
homogeneous

$$\lambda_0 a_{n+k} + \lambda_1 a_{n+k-1} + \lambda_2 a_{n+k-2} + \dots + \lambda_k a_{n+k-k} = f(n+k) = F(n)$$

$$\text{let } a_n = t^n$$

$$\therefore \lambda_0 t^{n+k} + \lambda_1 t^{n+k-1} + \lambda_2 t^{n+k-2} + \dots + \lambda_k t^n = F(n)$$

$$t^n [\lambda_0 t^k + \lambda_1 t^{k-1} + \lambda_2 t^{k-2} + \dots + \lambda_k] = F(n)$$

$$a_n [\lambda_0 t^k + \lambda_1 t^{k-1} + \lambda_2 t^{k-2} + \dots + \lambda_k] = F(n)$$

$$\lambda_0 t^k + \lambda_1 t^{k-1} + \lambda_2 t^{k-2} + \dots + \lambda_k = 0$$



Topic : Method of characteristic roots

Characteristic Equation : -

→ $\phi(E)$ is a Polynomial of degree = K
and let, $\phi(t) = 0$ is Characteristic Equation.
then Roots of Characteristic Equation are called Characteristic roots

let, $t_1, t_2, t_3, \dots, t_k$ are the Characteristic roots.

↑ 'k' Characteristic roots w.r.t. Polynomial of degree = K

* Complementary Function (C.F.) $\rightarrow$

$(a_n^{(H)})$ Complementary function is the solution w.r.t. homogeneous part of the linear recurrence relation
{i.e. w.r.t. $f(n) = 0$ }

* Particular Solution (P.S.)

$(a_n^{(P)})$ Particular solution is the solution w.r.t. non-homogeneous part of linear recurrence relation
{i.e. when $f(n) \neq 0$ }

* Complete Solution:-

Complete Solution of linear recurrence relation = (Complementary function + Particular solution)
$$a_n = a_n^{(H)} + a_n^{(P)}$$

Complete Solution (a_n) : -

$$a_n = CF + P.S.$$

(or)

$$a_n = a_n^{(H)} + a_n^{(P)}$$

Note:-

If given recurrence relation is a homogeneous linear recurrence relation {i.e, if $f(n)=0$ },

then Complementary function alone will be the solution of that recurrence relation

Because,

When $f(n)=0$,
then Particular Solution = 0

Consider the following recurrence relation.

$$3a_n + 5a_{n-1} + a_{n-2} = 2^n$$

let $a_n^{(H)}$ is the solution for homogeneous part

$$\text{i.e., } 3a_n^{(H)} + 5a_{n-1}^{(H)} + a_{n-2}^{(H)} = 0$$

let $a_n^{(P)}$ is the solution w.r.t. non-homogeneous part

$$\text{i.e., } 3a_n^{(P)} + 5a_{n-1}^{(P)} + a_{n-2}^{(P)} = 2^n$$

Complete solution is $a_n = (a_n^{(H)} + a_n^{(P)})$

$$\text{L.H.S} = 3a_n + 5a_{n-1} + a_{n-2}$$

$$= 3(a_n^{(H)} + a_n^{(P)}) + 5(a_{n-1}^{(H)} + a_{n-1}^{(P)}) + (a_{n-2}^{(H)} + a_{n-2}^{(P)})$$

$$= \underbrace{(3a_n^{(H)} + 5a_{n-1}^{(H)} + a_{n-2}^{(H)})}_0 + \underbrace{(3a_n^{(P)} + 5a_{n-1}^{(P)} + a_{n-2}^{(P)})}_{2^n}$$

$$= 2^n = \underline{\text{R.H.S.}}$$

Rules to write Complementary Function

Characteristic Roots

Complementary function

*① When all roots are real and distinct.

ie; $t_1, t_2, t_3, \dots, t_k$

$$\textcircled{1} C_1 \cdot (t_1)^n + C_2 \cdot (t_2)^n + \dots + C_k (t_k)^n$$

Where $C_1, C_2, C_3, \dots, C_k$ are Unknown Constants

② When all roots are real and two roots are equal

ie; $t_1, t_1, t_3, t_4, \dots, t_k$

$$\textcircled{2} (C_1 + C_2 \cdot n) \cdot (t_1)^n + C_3 (t_3)^n + C_4 (t_4)^n + \dots + C_k (t_k)^n$$

Where $C_1, C_2, C_3, \dots, C_k$ are Unknown Constants

③ When all roots are real and three roots are equal

ie; $t_1, t_1, t_1, t_4, \dots, t_k$

$$\textcircled{3} (C_1 + C_2 n + C_3 n^2) \cdot (t_1)^n + C_4 (t_4)^n + \dots + C_k (t_k)^n$$

Where $C_1, C_2, C_3, \dots, C_k$ are Unknown Constants

* Note:- We can write the Complementary Function
Even when some roots are complex
{ Not required wrt. GATE }

Note:-

① In equation ① if $f(n) = 0$, then recurrence relation will be homogeneous linear recurrence relation, Complementary function alone will be the complete solution of the given recurrence relⁿ
i.e., if $f(n) = 0$, then $A_n = C.F + P.S.$
 $A_n = C.F$

② In equation ① if $f(n) \neq 0$, then recurrence relation will be non-homogeneous linear recurrence relation, and complete solution can be given by
 $A_n = C.F + P.S.$ { i.e., if $f(n) \neq 0$, then we need to obtain the particular solution }

Particular Solution

Particular Solution :-

→ From Eqⁿ ③

$$a_n \cdot \phi(E) = F(n) \quad \text{--- eqⁿ ③}$$

$$a_n = \left(\frac{F(n)}{\phi(E)} \right)$$

Particular Solution ($a_n^{(P)}$) is the solution of $\frac{F(n)}{\phi(E)}$

$$\text{P.S.} = (a_n^{(P)}) = \text{Solution of } \left(\frac{F(n)}{\phi(E)} \right)$$

after replacing n by ' $n+k$ '
in $f(n)$

If $f(n) = 0$

then $F(n) = 0$

∴ in $a_n \cdot \phi(E) = F(n)$

$$a_n = \frac{F(n)}{\phi(E)}$$

Particular
Solun = Solun of $\left(\frac{F(n)}{\phi(E)} \right)$

= Solution of $\left(\frac{0}{\phi(E)} \right)$

= 0

Rules for writing Particular Solution When $F(n) = (b)^n$

"Constant"

Exponential
function
with base 'b'

$$P.S = \text{Soln of } \left(\frac{F(n)}{\phi(E)} \right) = \text{Soln of } \left(\frac{b^n}{\phi(E)} \right)$$

Case ① When 'b' is not a characteristic root [i.e. when $\phi(b) \neq 0$]
in that case

$$\text{Particular solution} = \frac{(b)^n}{\phi(b)}, \text{ when } \phi(b) \neq 0$$

$(a_n^{(p)})$

Rules for writing Particular Solution When $F(n) = (b)^n$ Exponential function with base 'b'

$$P.S = \text{Soln of } \left(\frac{F(n)}{\phi(E)} \right) = \text{Soln of } \left(\frac{b^n}{\phi(E)} \right)$$

Case ② :- When ' b ' is a characteristic root with multiplicity ' m '.

i.e., $\phi(b) = 0$ and $\phi(E) = (E-b)^m \psi(E)$ {when $\psi(b) \neq 0$ }

in that case

$$\text{Particular Solution} = Q_n^{(p)} = \text{Solution of } \left(\frac{(b)^n}{\phi(E)} \right) = \text{Solution of } \left(\frac{(b)^n}{(E-b)^m \psi(E)} \right) = \frac{1}{\psi(b)} \cdot {}^nC_m \cdot (b)^{n-m}$$

$$= \frac{1}{\psi(b)} \cdot {}^nC_m \cdot (b)^{n-m}$$

Q:- Find the solution of the recurrence relation,
Where $a_0 = 1$
 $a_n = 3 \cdot a_{n-1}$

Soln:

$$a_n = 3 \cdot a_{n-1}$$

$$a_n - 3a_{n-1} = 0$$

homogeneous
linear
recurrence
relation

Put $n-1 = n$, i.e. $n = n+1$

∴ Equation becomes

$$a_{n+1} - 3a_n = 0$$

Use shift operator. $a_{n+1} = E^1 a_n$
∴ Eqⁿ becomes

$$E^1 a_n - 3a_n = 0$$

$$a_n (E - 3) = 0$$

$\phi(E)$ $\leftarrow$ F(n)

$$\phi(E) = E - 3$$

∴ Ch. eqⁿ is

$$\phi(t) = 0$$

$$\text{i.e. } (t - 3) = 0$$

it gives $t = 3$

∴ Ch. root is $t = 3$

Hence Complementary
function (C.F) is.

$$C.F = C_1 \cdot (3)^n$$

Given recurrence relⁿ
is homogeneous linear
recurrence relation

$$\therefore \text{P.S.} = 0$$

→ Complete Soluⁿ

$$a_n = C.F. + \text{P.S.} \rightarrow 0$$

$$a_n = C_1 \cdot (3)^n$$

We know $a_0 = 1$

$$a_0 = C_1 \cdot (3)^0$$

$$1 = C_1 \cdot 1 \Rightarrow C_1 = 1$$

$$\text{Hence } a_n = 3^n$$

Q:- Find the solution of
 $a_n - 7a_{n-1} + 12a_{n-2} = 0$

Soln:

$$a_n - 7a_{n-1} + 12a_{n-2} = 0$$

Put $n-2=n$, i.e. $n=n+2$

∴ Eqⁿ becomes

$$a_{n+2} - 7a_{n+1} + 12a_n = 0$$

Use shift operator $a_{n+1} = E \cdot a_n$

∴ Eqⁿ becomes

$$E^2 a_n - 7 \cdot E \cdot a_n + 12 a_n = 0$$

$$a_n (E^2 - 7E + 12) = 0$$

$\phi(E)$

the recurrence relation,

Where $\boxed{a_0 = 1}$
 $\boxed{a_1 = 2}$

$$\phi(E) = E^2 - 7E + 12$$

∴ Ch eqⁿ is

$$t^2 - 7t + 12 = 0$$

$$\boxed{t = 3, 4}$$

Ch. roots, $t_1 = 3$ & $t_2 = 4$

Hence,

Complementary function

$$C.F. = C_1(3)^n + C_2(4)^n$$

Given recurrence relⁿ
is homogeneous
∴ P.S. = 0

Complete Soluⁿ.

$$a_n = C.F. + P.S. \rightarrow 0$$

$$a_n = C_1(3)^n + C_2(4)^n$$

$$a_0 = 1 \text{ \& } a_1 = 2$$

$$a_0 = 1 = C_1(3)^0 + C_2(4)^0$$
$$\Rightarrow \boxed{C_1 + C_2 = 1} \text{ --- (1)}$$

$$a_1 = 2 = C_1(3)^1 + C_2(4)^1$$
$$\Rightarrow \boxed{2 = 3C_1 + 4C_2} \text{ --- (2)}$$

By ① & ② $C_2 = -1$ & $C_1 = 2$

Hence

$$\boxed{a_n = 2(3)^n - 1(4)^n}$$

Q:- Find the solution of the recurrence relation,
 where $a_1 = 2$

$$a_n = 2a_{n-1} - 1$$

Soln:

$$a_n - 2a_{n-1} = -1 \quad \text{f(n)}$$

$$f(n) = -1 = -1 \cdot (1)^n$$

Put $n-1 = n$ i.e. $n = n+1$

$$a_{n+1} - 2a_n = -1$$

Using shift operator

$$E \cdot a_n - 2 \cdot a_n = -1$$

$$a_n (E-2) = -1 \cdot (1)^n$$

$\searrow \quad \nearrow$
 $\phi(E) \quad F(n)$

the recurrence relation,

where $a_1 = 2$

$$\phi(E) = E-2$$

$$\therefore \text{Ch eq} \Rightarrow t-2=0$$

Hence Ch. root $\Rightarrow t_1 = 2$

and

Complementary function

$$C.F = C_1 \cdot (2)^n$$

$$a_n (E-2) = -1 \cdot (1)^n$$

$$a_n = \frac{-1 \cdot (1)^n}{(E-2)} = - \frac{(1)^n}{\phi(E)}$$

P.S:- Soln of $\left\{ - \frac{(1)^n}{\phi(E)} \right\}$

$$= - \left\{ \text{Soln of } \frac{(1)^n}{\phi(E)} \right\}$$

base of Exponent = 1

'1' is not a Characteristic root $\therefore \text{Soln of } \left\{ \frac{(1)^n}{\phi(E)} \right\} = \frac{(1)^n}{\phi(1)}$

$$= - \left\{ \frac{(1)^n}{\phi(1)} \right\} = - \left\{ \frac{1}{(1-2)} \right\} = - \left\{ \frac{1}{-1} \right\} = +1$$

P.S = 1

Complete Solution

$$a_n = CF + P.S.$$

$$a_n = C_1 \cdot (2)^n + 1$$

given $a_1 = 2$

$$a_1 = 2 = C_1 \cdot (2)^1 + 1$$

$$\Rightarrow 2 = 2 \cdot C_1 + 1$$

$$\Rightarrow C_1 = 1/2$$

$$\therefore a_n = \frac{1}{2} \cdot (2)^n + 1$$

$$a_n = 2^{n-1} + 1$$

Ans

H.W.
Q:- Find the solution of the recurrence relation,
 $a_n = 2a_{n-1} + 2^n$ $a_0 = 1$

Soln:

HW: Q:- Find the soln of recurrence relation,

$$a_n - 2a_{n-1} + a_{n-2} = 0$$

$$\begin{array}{l} a_0 = 1 \\ a_1 = 2 \end{array}$$

H.W.

Q:- Find the solution of recurrence relation

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

$a_0 = 1$ $a_1 = 2$

HW Q: Find the solution of recurrence relation

$$T(3^k) = 2 \cdot T(3^{k-1}) + 1$$

$$\text{let } \boxed{T(1) = 1}$$

$$\text{let } T(3^k) = Q_k$$

$\therefore$ Eqⁿ becomes

$$\boxed{Q_k = 2 \cdot Q_{k-1} + 1}$$

↳ Now solve
the question



2 mins Summary



Topic

Solution of recurrence relation using method of characteristic roots

Slide

THANK - YOU